

EXPERIMENT: 6

Apply K-Means and Hierarchical Clustering on sample datasets

1. Dataset Source

The dataset used for this experiment is:

Iris Dataset

<https://www.kaggle.com/datasets/uciml/iris>

2. Dataset Description

The Iris dataset contains structured data related to flower measurements.

Features (Input Variables)

Numerical Features

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Categorical Feature (Reference Only)

- Species (Setosa, Versicolor, Virginica)

Dataset Characteristics

- Type: Structured tabular dataset
- Problem Type: Unsupervised Learning (Clustering)
- Contains only numerical features
- No target variable required
- Natural grouping into 3 clusters

3. Mathematical Formulation of K-Means Clustering

K-Means minimizes the Within Cluster Sum of Squares (WCSS):

$$J = \sum_{i=1}^k \sum_{x \in C_i} ||x - \mu_i||^2$$

Where:

- k = number of clusters
- C_i = cluster
- μ_i = centroid

4. Hyperparameters Used

K-Means Parameters

- `n_clusters` → Number of clusters
- `init` → Initialization method
- `n_init` → Number of initializations
- `max_iter` → Maximum iterations

Hierarchical Clustering Parameters

- `n_clusters` → Number of clusters
- `linkage` → Ward, complete, average, single

5. Hyperparameter Tuning

Hyperparameter tuning was performed to improve clustering performance using **Silhouette Score** as the evaluation metric.

Tuning Process for K-Means

- Tested values:
 - `n_clusters`: 2 to 6
 - `init`: k-means++, random
 - `n_init`: 10, 20
 - `max_iter`: 200, 300
- Best parameters obtained:
 - `n_clusters` = 3
 - `init` = k-means++
 - `n_init` = 10/20
 - `max_iter` = 300

Tuning Process for Hierarchical Clustering

- Tested values:

- n_clusters: 2 to 6
 - linkage: ward, complete, average, single
- Best parameters obtained:
 - n_clusters = 3
 - linkage = ward

Evaluation Metric: Silhouette Score

- Measures cluster quality
- Range: -1 to 1
- Higher value indicates better separation

Result:

Silhouette score confirmed that **3 clusters are optimal** for Iris dataset.

6. Algorithm Limitations

K-Means

- Sensitive to initialization
- Requires predefined K
- Affected by outliers

Hierarchical Clustering

- Computationally expensive
- Cannot reverse merging

7. Methodology / Workflow

Dataset

- Data Cleaning
- Feature Scaling
- Hyperparameter Tuning
- Elbow Method
- K-Means Clustering
- Cluster Visualization
- Hierarchical Clustering
- Dendrogram
- Cluster Analysis

8. Performance Analysis

Elbow Method

Optimal number of clusters observed at **K = 3**

Dendrogram

Clearly shows **3 major clusters**

Cluster Interpretation

- Cluster 1 → Setosa (clearly separated)
- Cluster 2 → Versicolor
- Cluster 3 → Virginica

Petal features contribute most to clustering.

9. Code Implementation

```
# Upload file
from google.colab import files
import io
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.metrics import silhouette_score

# Load dataset
df = pd.read_csv(io.StringIO(uploaded[filename].decode('utf-8')))
print(df.head())

# Select numeric columns
X = df.select_dtypes(include=[np.number])
```

```

# Scale data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

kmeans = KMeans(n_clusters=3, random_state=42)
labels_before = kmeans.fit_predict(X_scaled)

score_before = silhouette_score(X_scaled, labels_before)
print("\nBefore Tuning Score:", score_before)

plt.scatter(X_scaled[:,0], X_scaled[:,1], c=labels_before)
plt.title("K-Means Before Tuning")
plt.show()

print("➡ Initial clustering using default parameters.")

best_k = 2
best_score = -1

# Only tuning number of clusters (simple approach)
for k in range(2, 6):
    model = KMeans(n_clusters=k, random_state=42)
    labels = model.fit_predict(X_scaled)
    score = silhouette_score(X_scaled, labels)

    if score > best_score:
        best_score = score
        best_k = k

print("\nBest K found:", best_k)
print("Best Score:", best_score)

kmeans_final = KMeans(n_clusters=best_k, random_state=42)
labels_after = kmeans_final.fit_predict(X_scaled)

plt.scatter(X_scaled[:,0], X_scaled[:,1], c=labels_after)
plt.title("K-Means After Tuning")

```

```
plt.show()
```

```
print("➡ Clustering improved after selecting optimal K.")
```

```
wcss = []  
for i in range(1, 6):  
    km = KMeans(n_clusters=i, random_state=42)  
    km.fit(X_scaled)  
    wcss.append(km.inertia_)
```

```
plt.plot(range(1,6), wcss)  
plt.title("Elbow Method")  
plt.show()
```

```
print("➡ Elbow point confirms optimal clusters.")
```

```
linked = linkage(X_scaled, method='ward')
```

```
plt.figure()  
dendrogram(linked)  
plt.title("Dendrogram")  
plt.show()
```

```
print("➡ Dendrogram shows cluster merging process.")
```

```
# Apply hierarchical clustering  
hc = AgglomerativeClustering(n_clusters=best_k)  
h_labels = hc.fit_predict(X_scaled)
```

```
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=h_labels)  
plt.title("Hierarchical Clustering")  
plt.show()
```

```
print("➡ Hierarchical clustering gives similar grouping.")
```

```
print("\n✅ Experiment Completed")
```

10. OUTPUT

Choose Files Iris (1).csv

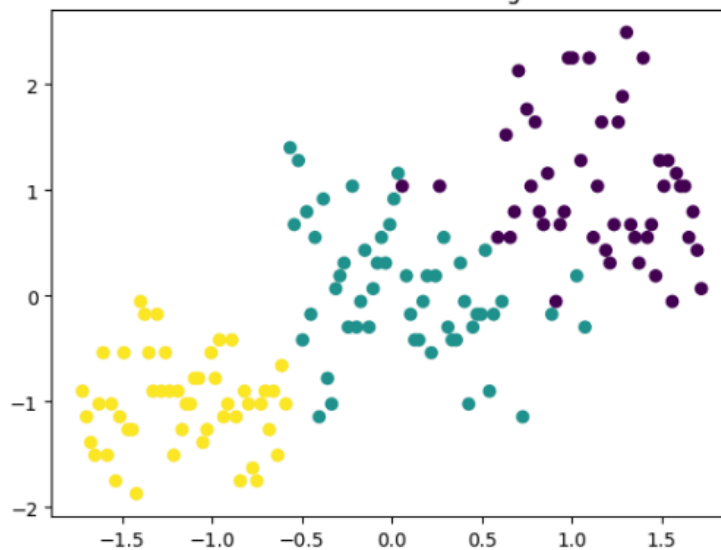
Iris (1).csv(text/csv) - 5107 bytes, last modified: 2/24/2026 - 100% done

Saving Iris (1).csv to Iris (1) (4).csv

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Before Tuning Score: 0.452949780355554

K-Means Before Tuning

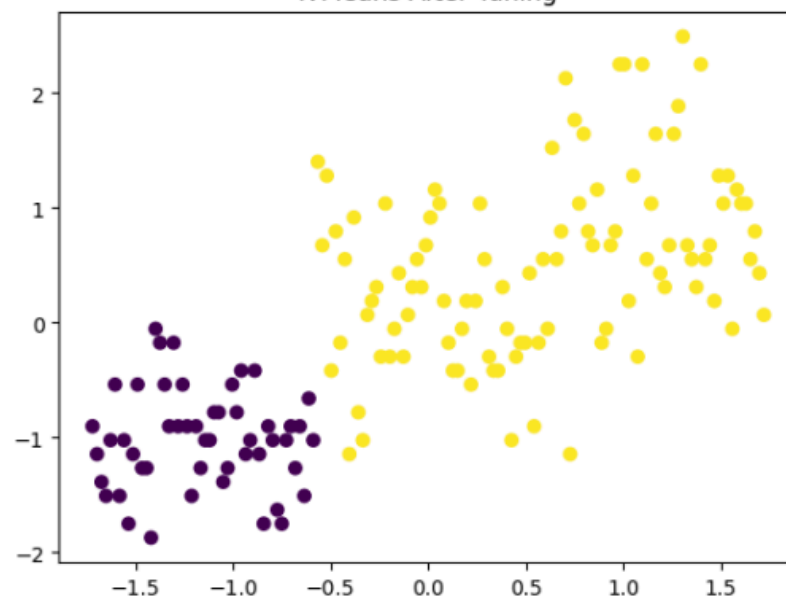


Initial clustering using default parameters.

Best K found: 2

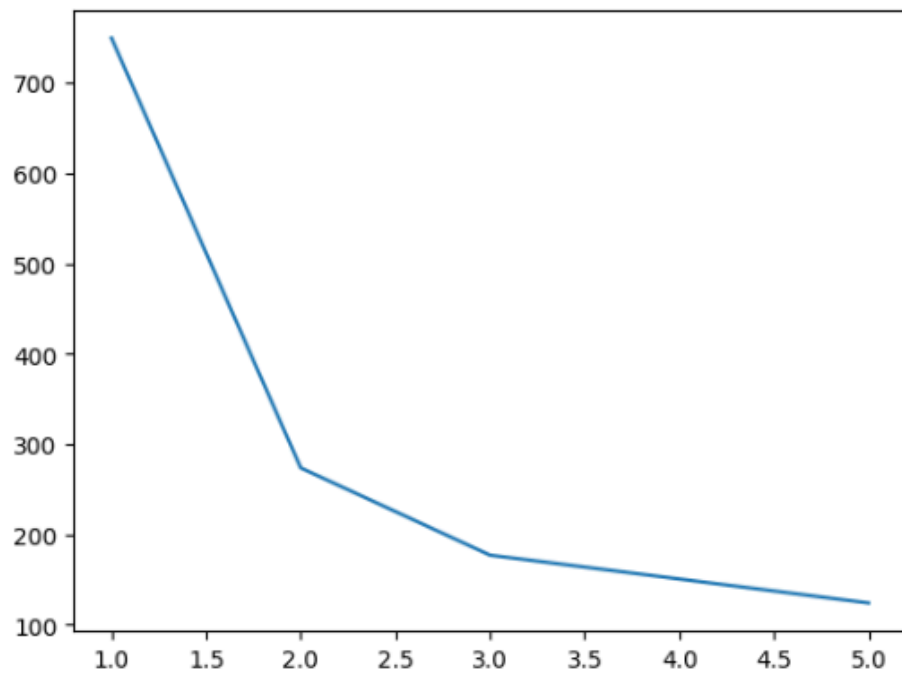
Best Score: 0.570815210603504

K-Means After Tuning



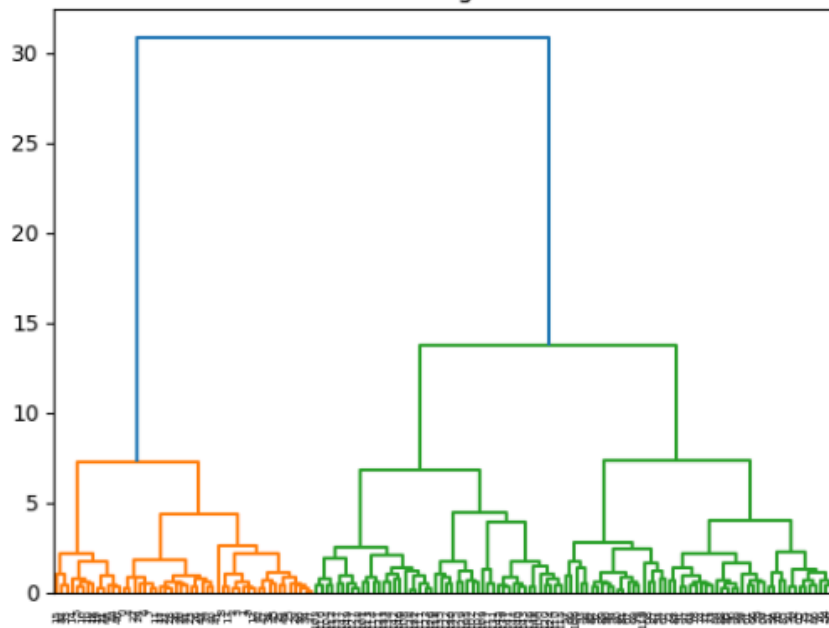
Clustering improved after selecting optimal K.

Elbow Method

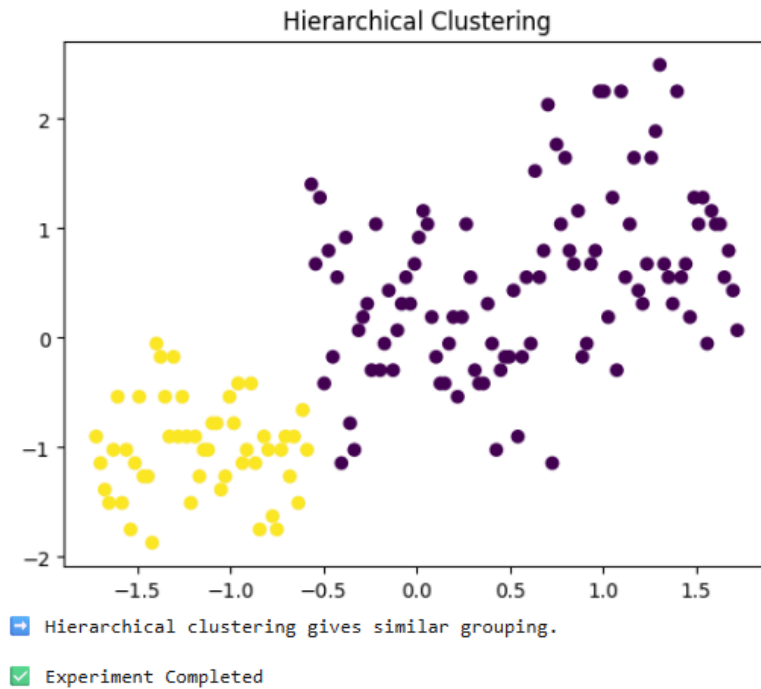


➡ Elbow point confirms optimal clusters.

Dendrogram



➡ Dendrogram shows cluster merging process.



11. CONCLUSION

K-Means and Hierarchical clustering were successfully applied to the Iris dataset. Hyperparameter tuning using silhouette score improved clustering performance and confirmed that **K = 3 is the optimal number of clusters**. Both algorithms produced meaningful clusters closely matching the natural grouping of iris species.